Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 · Functional materials and green chemistry · Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 1)

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
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Low ASR confidence	Transcript segment requires audio verification.
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M1

Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 1)

1 Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 1)

Study focus: Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 1). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000018 · EV0000019

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will start with a new module today on nanomaterials.

Key details

- The content will be nanomaterials, classification and deviation of properties from bulk material, and we will look at different properties of nanoparticles.
- In today's class we will study of nanomaterials, about nanomaterials, the classification of nanoparticles, and the reasons for deviation in properties of nanomaterials from the bulk material.
- The prefix nano is derived from the Greek word dwarf meaning very tiny.
- One nanometer would be approximately equal when we have ten hydrogen atoms placed one next to the other or five silicon atoms.
- aligned in a line.
- Previous module we studied about molecular spectroscopy and we understood that when we study about molecules or atoms we need to use the quantum mechanical treatment.
- But when the dimension becomes close to the nanometer, then we are going to observe.
- Because the dimension are now, dimensions are closer to those of atoms or molecules.
- because their properties are very different from that of the bulk materials.

- All of us know that bulk gold as we see it is yellowish in colour.
- That means there is some change in the optical property.
- That means the activity has become so high something involved which is passive has become so active that it is combustible to the extent that it is used as a solid fuel in rocket propulsion.
- Another property of importance is melting point of the materials when they are taken in the nanoparticle range becomes lower.
- If we want to check purity of the material, we always quote the melting point.
- This material will be classified as a nanomaterial.
- And what would be nano science?
- It is the study of the phenomena and synthesis, characterization, exploration, and exploitation of these nanostructure materials.

Example or application from the lecture

- Let us first understand what we mean by a nanoparticle.
- For example, gold.
- In fact the a name golden colour has been given to gold.

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Core explanation · 05:00–09:59

Before studying the reasons why nanomaterials behave so differently from the bulk material, let us talk about classification of nanomaterials based on their dimensions.

Key details

- Now the first classification would be it will be called as a zero dimensional nanomaterial.
- What does it mean?
- Range exceeding 100 nanometers.
- It's understood now that two of the dimensions are allowed to be outside of the nanorange.
- films these will be classified as two dimensional nanomaterials only one dimension is now within the nano range but still we call it as a nanomaterial.
- And then coming to three dimensional nanomaterials where all the three dimensions are outside of the nano range.
- Then how is it a nanomaterial only specific types of materials where you have like say bundles of nanovires.
- Nanomaterial, therefore, these overall will look like as if all the three dimensions are outside of the nanorange.
- First and foremost reason which we should discuss here is that the large fraction of surface atoms per unit volume.
- Atoms on the surface, but major of the atoms are in the bulk of the material.
- But when we come to nanomaterials, what happens?
- The particle size has.

- Become so small now that on the surface you have a large number of surface atoms, whereas in the bulk we have very few atoms now.
- Properties now are strictly governed by the surface atom because they are larger in number as compared to the bulk atoms.
- When we have a platinum sheet as compared, which is we can say is bulk, whereas compared to platinum powder, but to a nanopowder.
- Now the surface area, each particle now, the surface area has is more, and therefore overall surface area.

Example or application from the lecture

- Example, as I said, nano clusters or quantum dots fall into this category of nanomaterials.
- An example would be nanotubes and nanovires we can understand look at the picture here.
- So such materials are also classified as three dimensional nanomaterials where we have all the three dimensions outside of the nanorange like for example dispersions of nanoparticles or we have bundles of nanovires.

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Core explanation · 10:00–14:59

has become so high that the number of surface atoms are very large and therefore we will find that the properties now will be dictated by these atoms rather than the bulk atoms.

Key details

- We have a box like this.
- Dividing it further and further, the surface area is definitely going to increase and the number of surface atoms will go on increasing.
- We all know that the bulk atoms are surrounded by atoms on all sides and therefore there's a force of attraction bit and repulsion between these bulk atoms.
- But when we look at the surface atoms they are being held or pulled inside or held together with only one on only one side that other side other side is exposed to the environment therefore there is they are held with less force and therefore there is more energy on the surface and when any.
- System has more energy tries to it tries to reduce its energy and therefore we'll find that the surface activity will be very high.
- Of nanoparticles, we find that the same aluminium becomes so combustible that it is being used as rocket fuel.
- So therefore this.
- The reason being that there are large number of surface atoms and they are being held with less force, so surface energy is very high.
- Another reason which is very important and if we understand this, all properties, especially electrical and optical properties, can be explained very easily.
- by the use of the concept of spatial confinement or quantum confinement.
- We have these metal at the lattice points and we have a sea of electrons.

- That is for a bulk.
- You make it so small and so thin that it has now become into a two dimensional material.
- What would happen?
- when you make it into a two dimensional quantum well say.
- Your electrons can move only in one direction.
- In the other two directions, they are completely confined.
- You have changed, you have made it so thin that now it is allowed to move only in one di dimension.
- In the holes to move only in one direction, and when you come to zero dimension, which are called quantum dots, you have restricted the motion completely, and it's almost like within a very small area, you are confining.
- The electron and the poles.
- When we have a bulk material, we know that.
- Say it is one hole of the material or the metal we are considering.
- One hole would mean to the 10 to the power of 23 atoms.

Example or application from the lecture

- And just now I gave an example of aluminium where we find that aluminium in bulk is passive in nature, whereas when it is taken in the form of.
- We know that let us take the example of a metal.
- But now suppose you make this bulk material so thin, so thin, so thin.

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Core explanation · 15:00–19:59

compared to nanomaterials where we'll have a maximum of ten to thousand atoms.

Key details

- Now here, when we say one mole, that means we have 10 to the power 23 atoms, that means so many electrons, and also means that there are so many energy levels.
- They'll be very close and almost continuous.
- They will be continuous to the extent that in metals we know that the balance band and the conductance band are overlapping.
- When we come to semiconductors, of course there is a small gap, nevertheless, these levels and the levels in the valence band and the condensed band, there are so many levels because there are so many atoms.
- And these levels will be very close to one another that they will be continuous.
- But when we come to nano range, what happens is now we have reduced the number of atoms.
- There are fewer atoms, fewer electrons, so the density of states falls down.
- And what we observe, say we are in a two dimensional.
- quantum well then we'll find that the energy levels now because there are less number of atoms, less number of electrons, less number of energy stays.

- And then when we come to zero dimensional nanomaterial the energy levels now are completely discrete and the energy gap has increased.
- If you compare them with a molecule in a molecule we know we have absolutely discrete energy levels.
- to the lumino, the highest occupied molecular orbital to the lowest unoccupied molecular orbital, and this is the energy gap.
- But as these become more and more in number, if you go this way now.
- When we come to bulk, we find that the valence band, there are so many.
- Many energy levels now because of so many atoms there.
- or the spatiment.
- What do we understand?
- That due to spatient or quantum confinement, the energy levels become more and more discrete and the energy gap increases.
- This is the result of quantum confinement, and this is going to have a strong bearing on the properties, especially the electronic properties, electrical properties, and the optical properties.
- And one more reason.
- for the bulk material behaving differently from the nanomaterial, but the nanomaterial behaves different from the bulk, is that in the bulk material we find that there are lattice imperfections like wherever there are dislocations, kinks, etcetera, are found in the lattice.
- Because the metal lattice is formed in such a way that layer after layer after layer is getting deposited of the metal.
- And many times we do not get a perfect layer.
- Rather there are some imperfections which are like dislocations or kinks are possible.
- in the metal structure, it's a bulk metal structure.
- But when you come to the nanomaterial range, what has happened now you just have to they're very small in size the particles.
- Therefore very small lattice has to be formed and therefore there is less possibility of imperfections.

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Core explanation · 20:00–24:59

And in the next class we are going to study how exactly these reasons are affecting the thermal properties, optical properties and so on so forth of the nanomaterials.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9969; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00008	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

C Revision and retrieval practice